Part 4: Preliminary Results and Methods

Data and Scope

This prototype analyzes weather–price relationships in PJM. The export has 1,829 records across 49 nodes and 7 regions, covering 2021–2025 (earlyAI/recentAI), with target lmp and weather variables CDH, HDH, TEMP_C, RH_PCT, WIND_MS, DEW_C, SLP_HPA, and PRECIP_MM.

Methods

Results were generated by the following pipeline:

1. Ingest PJM day-ahead LMP/load data and node-region metadata.
2. Pull NOAA ISD hourly weather and map each node to a nearest station.
3. Align all sources to hourly timestamps and compute derived weather features.
4. Compute node-year weather-to-LMP correlations; aggregate by region and period.
5. Export JSON to `data/exports/` and sync to `public/data/` with `npm run sync:data`.

Key scripts used: `scripts/export_pjm_metadata.py`, `scripts/noaa_isd_pull.py`, and analysis-export scripts in `scripts/`.

Preliminary Results

Fig. **??** is the primary comparison of correlation shifts (`recentAI - earlyAI`) for data-center-heavy vs non-heavy nodes.

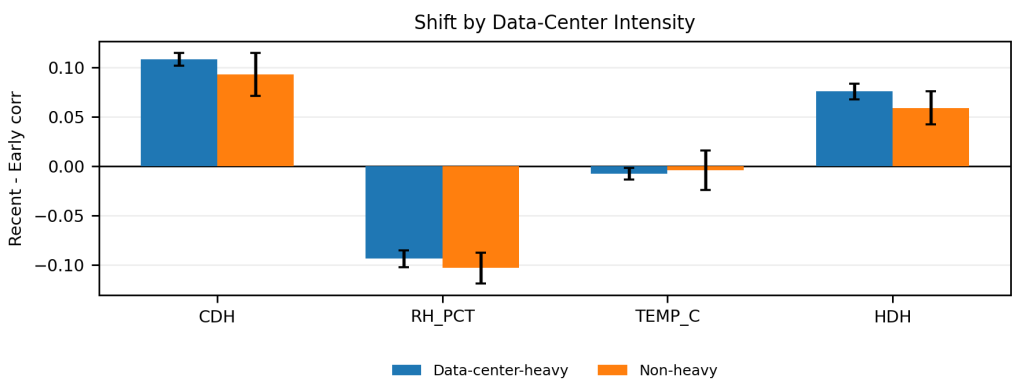

Figure 1: Mean shift in weather-to-LMP correlation (`recentAI - earlyAI`) by data-center intensity bucket, with standard-error bars.

Figure **??** represents how the weather–LMP relationship changed between periods for each bucket. Positive values indicate stronger recent correlation; negative values indicate weaker recent correlation. In our preliminary sample, CDH and HDH increase for both buckets (slightly more in the heavy bucket), while RH_PCT becomes more negative for both buckets (slightly more in the

non-heavy bucket). This indicates measurable temporal shifts, but not a uniform "heavy regions always weaken more" effect.

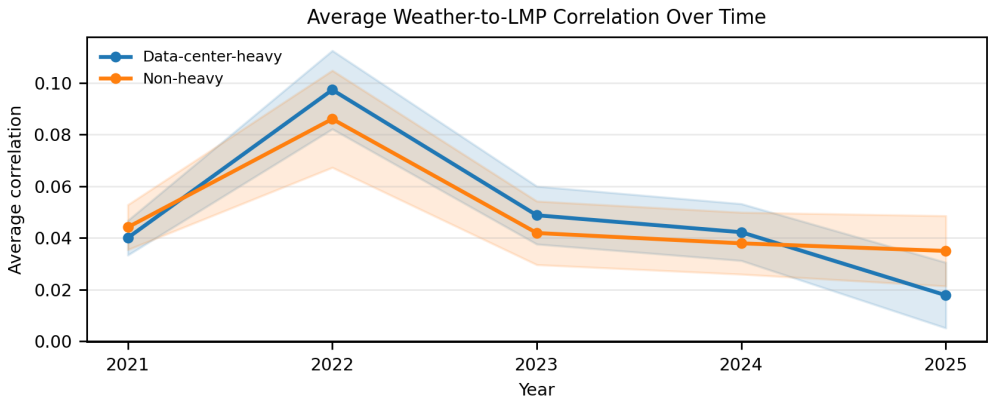

Figure 2: Supplemental trend view: yearly mean weather-to-LMP correlation by bucket (shaded bands are standard error).

As a supplement, Fig. **??** shows that both groups follow a similar time pattern (rise to 2022, then decline through 2025), with only small year-to-year gaps between heavy and non-heavy nodes.

Significance to Research Question

Our research question asks whether data-center-heavy areas exhibit stronger or systematically different weather–price relationships than non-heavy areas. The primary result (Fig. **??**) provides partial evidence of difference by variable (e.g., slightly larger CDH/HDH increases in heavy nodes), but not a consistent directional gap across all variables. The supplemental time-series result (Fig. **??**) reinforces this interpretation: both groups move similarly over time, suggesting common system-wide effects, with only modest group separation. Together, these preliminary findings support the project's central hypothesis that weather sensitivity is changing over time, while also showing that the "data-center-heavy vs non-heavy" distinction is nuanced and requires deeper modeling in later sections.

Implementation Access

Code repository: `https://github.com/jaiparmar940/energy-weather-app`
Data bundle (PJM + features): `https://duke.box.com/s/xtdrmjv1463wf0o60i1rmts0g0urxp01`

Reproduction (Minimal)

1. Clone repo; install dependencies.
2. Download shared PJM bundle to `data/pjm data` (or set `PJM_DATA_ROOT`).
3. Run data prep/analysis scripts and export JSON files.
4. Run `npm run sync:data`, then `npm run dev`.

Current Status

The analysis is preliminary (correlational, not causal), but the pipeline is functional and already shows nontrivial period differences across regions. Next steps are robustness checks (season/bucket), predictive baselines, and uncertainty reporting.